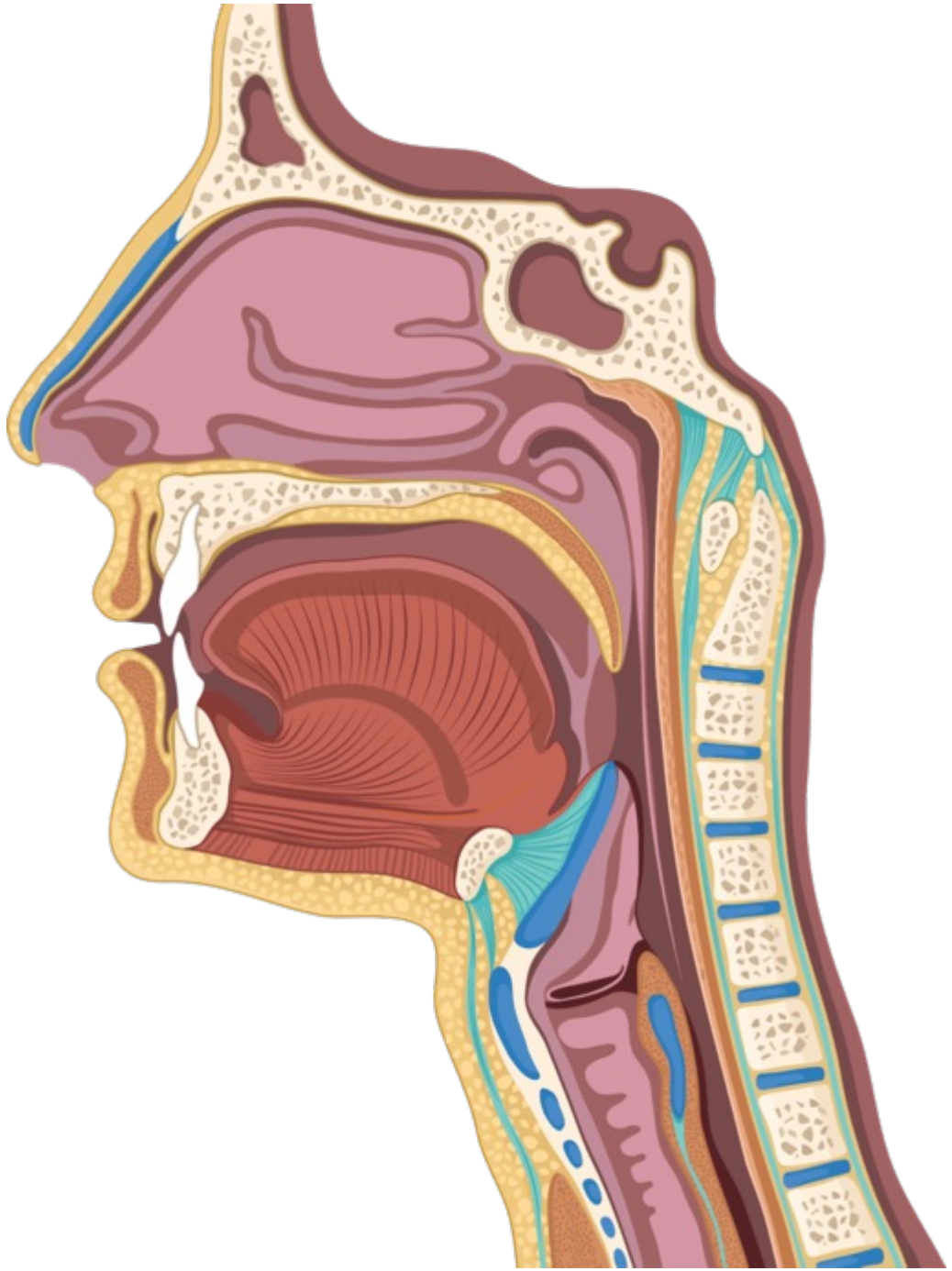
Guy's and St Thomas' NHS Foundation Trust

Royal Brompton & Harefield NHS Foundation Trust

Upper Respiratory Tract



Beccy Hall – Highly Specialist Cardiorespiratory Physiotherapist

With thanks to Isobel Davies for original slides

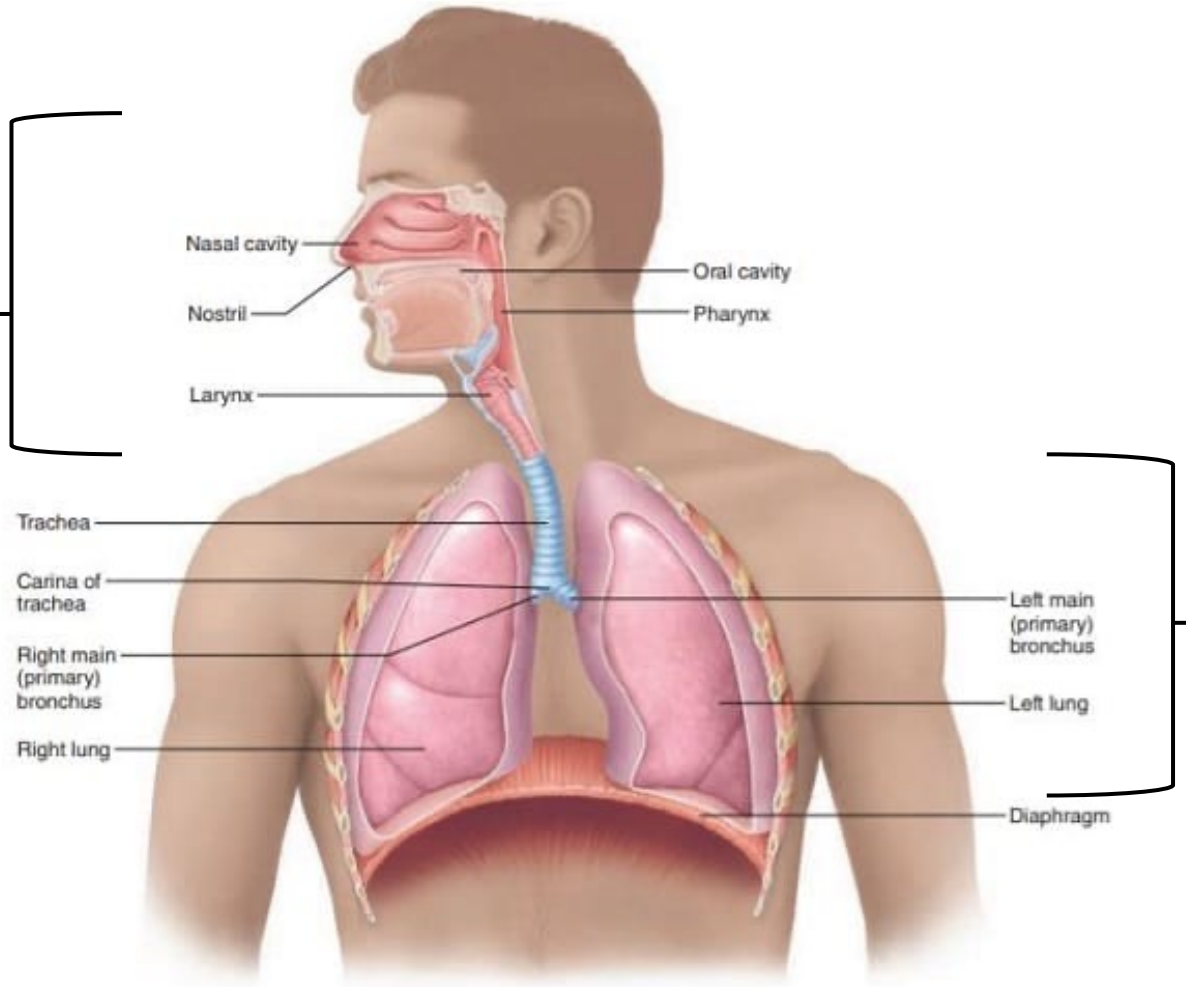
LONDON METROPOLITAN UNIVERSITY

Learning outcomes

- Understand the anatomy of the upper and lower respiratory tract
- Learn about the function and anatomy of the nasal turbines and paranasal sinuses
- Learn about the conducting and respiratory zones
- Understand the protective mechanisms of the lungs
- Learn about the cough reflex

Anatomy of the respiratory tract

Upper respiratory tract
Nose
Nasal cavity
Pharynx
Larynx



Lower respiratory tract
Trachea
Primary bronchi
Lungs

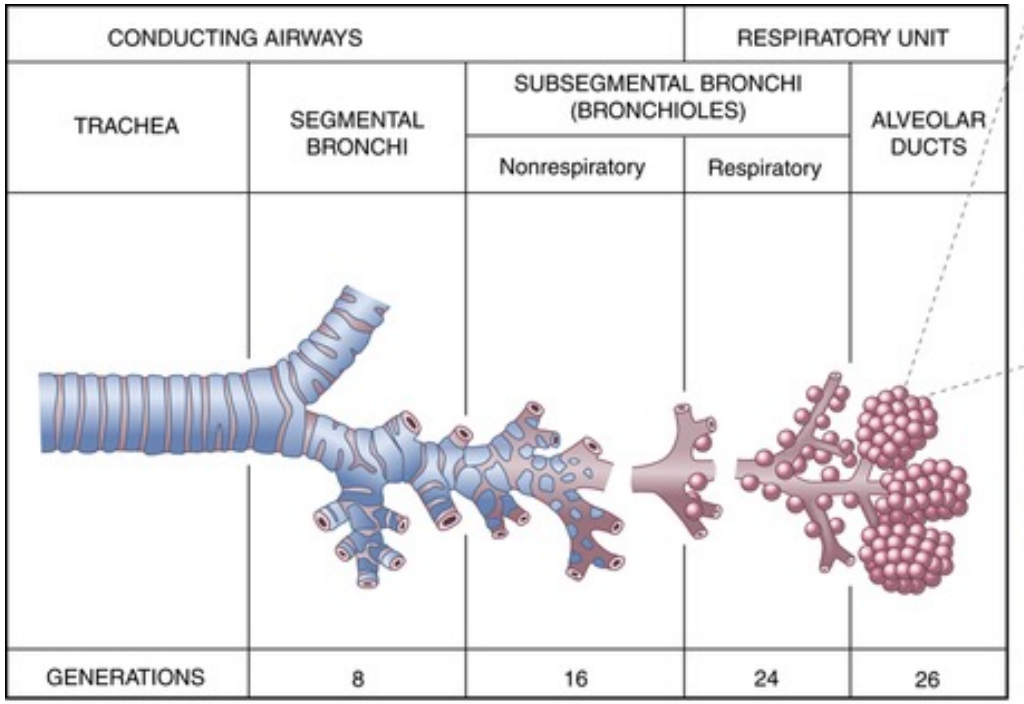
Conducting vs Respiratory zone

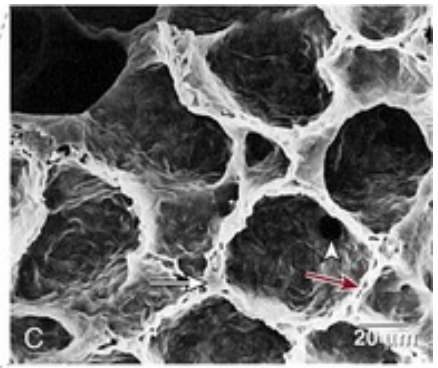
Conducting Zone:

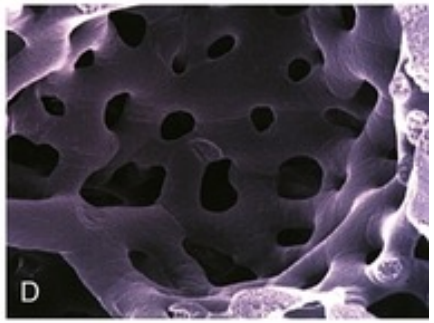
- Made up of the nose, pharynx, larynx , trachea, bronchi and bronchioles.
- Form a continuous passageway for air to move in and out of the lungs

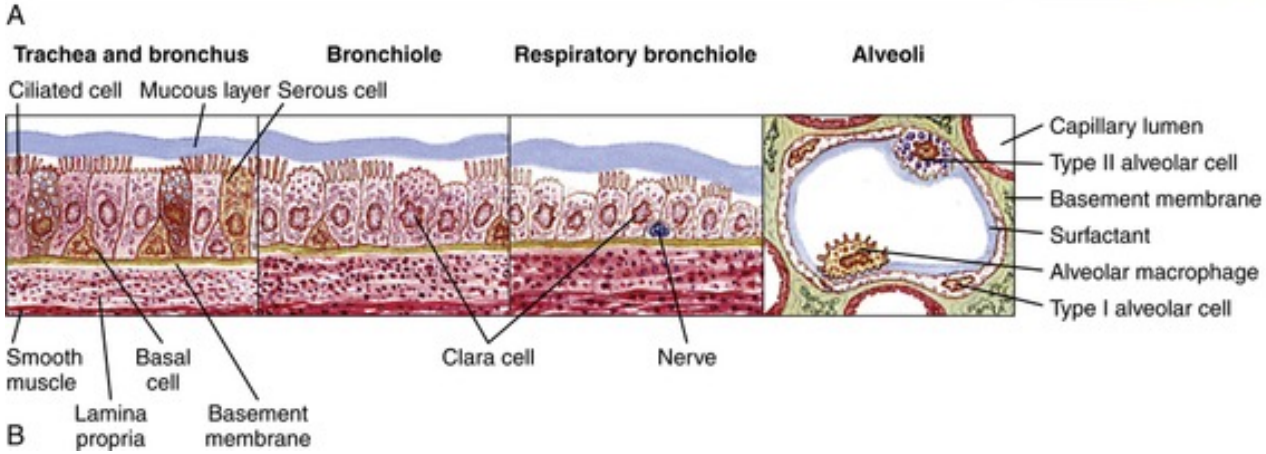
Respiratory Zone:

- Deep inside the lungs
- Respiratory bronchioles, alveolar ducts and alveoli









Anatomy of the upper respiratory tract

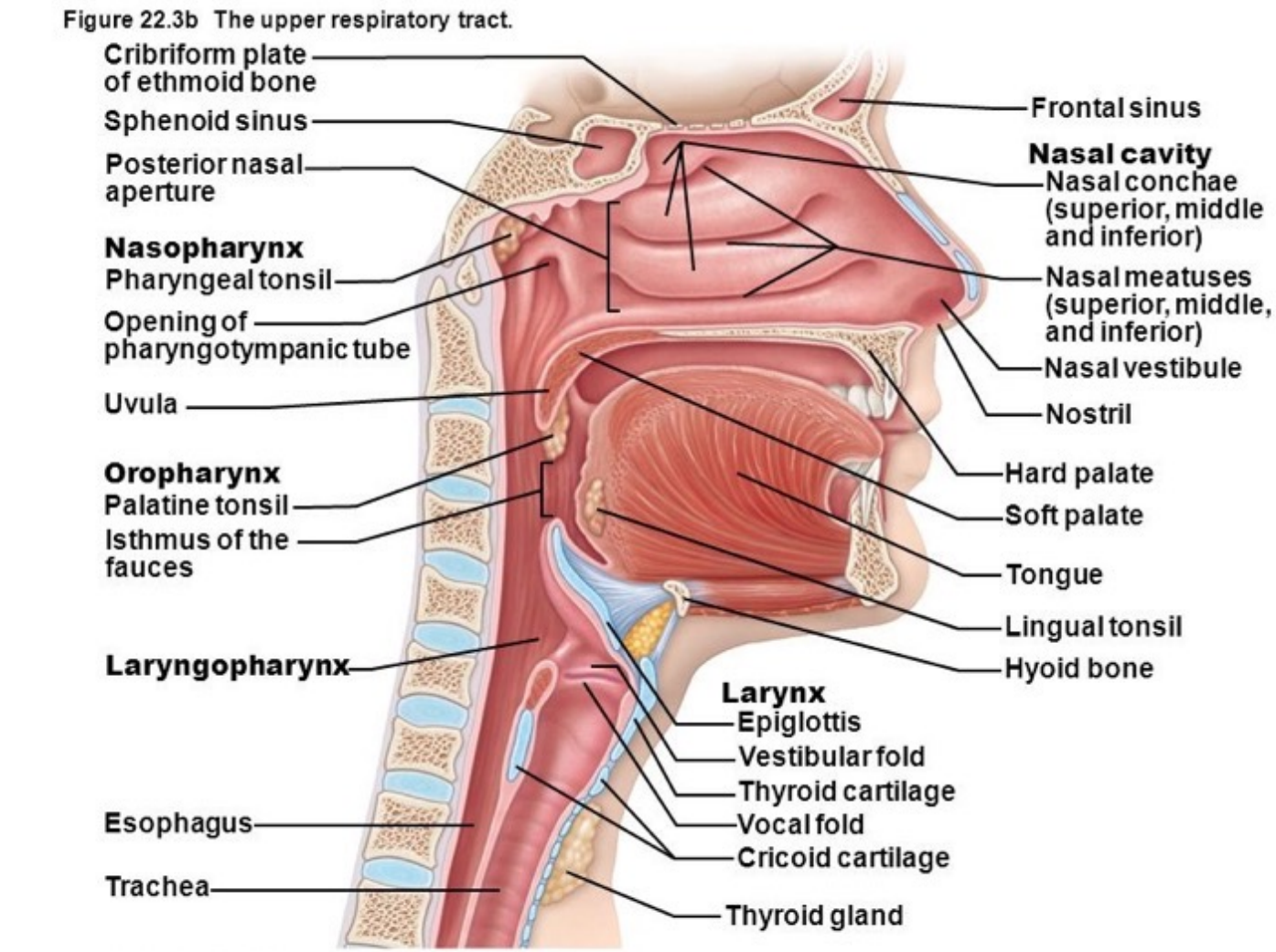


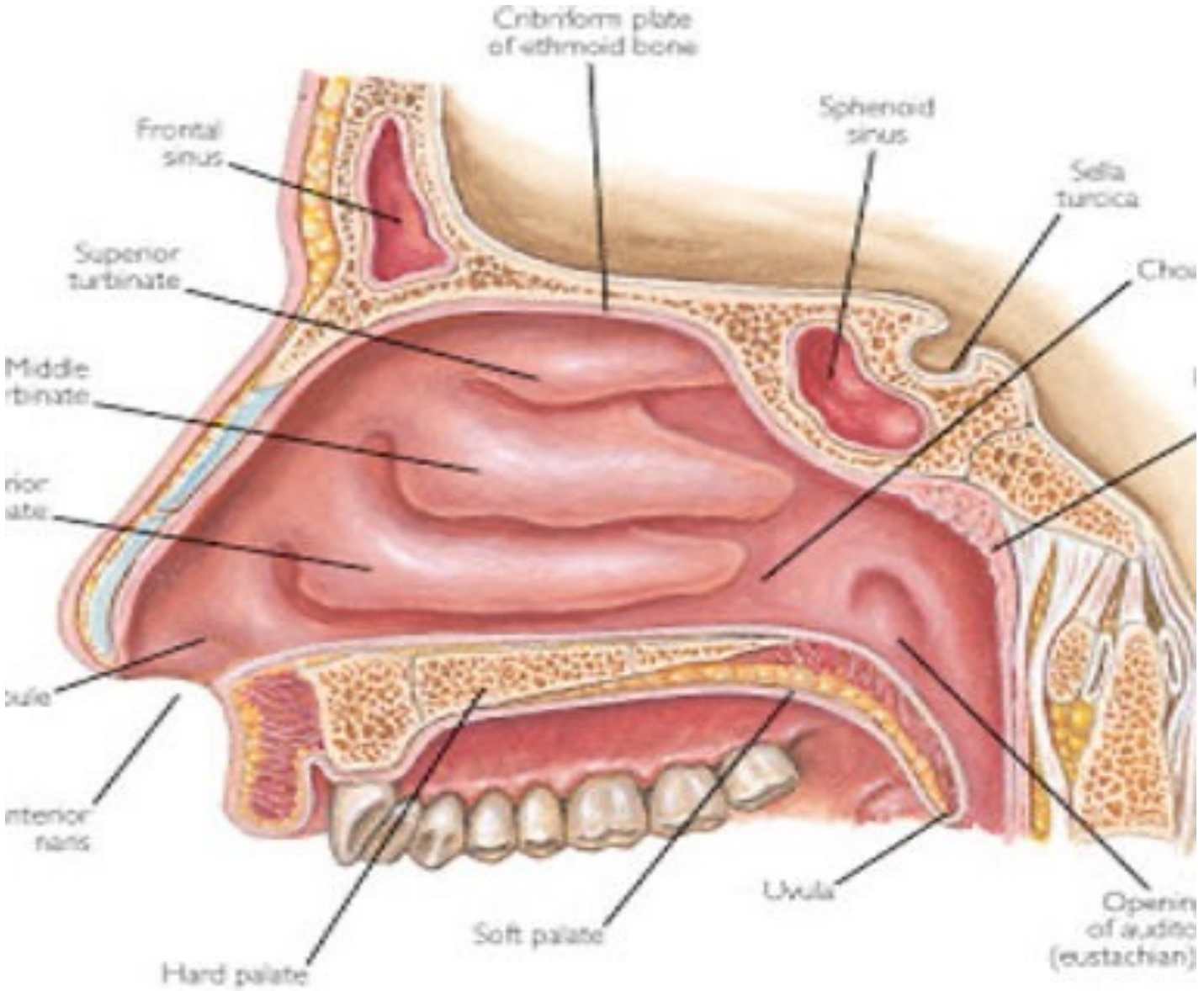
Figure 22.3b The upper respiratory tract.

(b) Illustration

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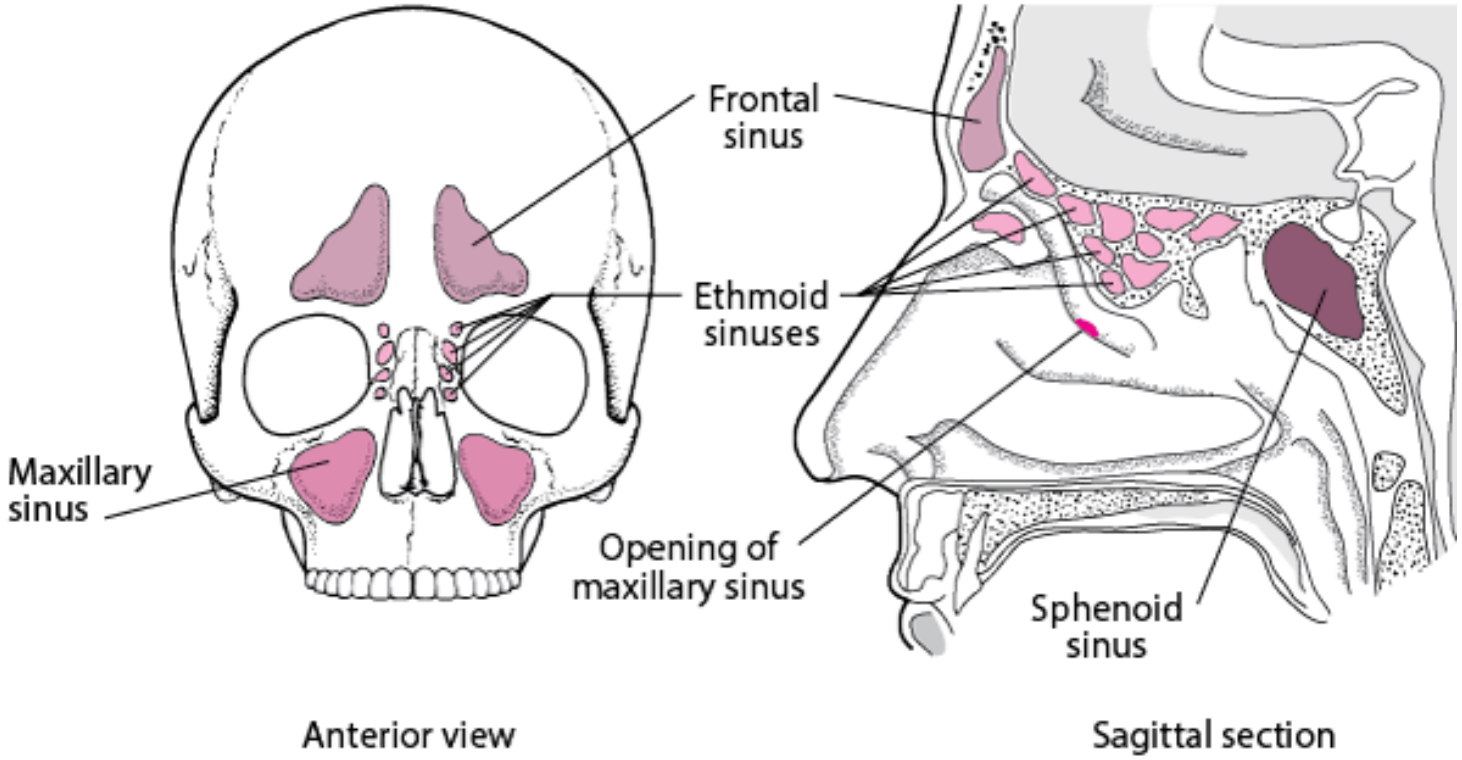
Nasal Turbinates

- Curved shelves that protrude from the lateral nasal walls
- Increase the surface area
- Responsible for warming and humidifying the air as it passes through the nasal cavity and into the nasopharynx and on to the lungs
- Also play a role in immune surveillance and olfaction



Paranasal Sinuses

- Decrease the relative weight of the bones of the face
- Increase the resonance of the voice
- Provide a buffer against blows to the face
- Insulate sensitive structures like dental roots and eyes from rapid temperature fluctuations in the nasal cavity
- Humidification and heating of inhaled air

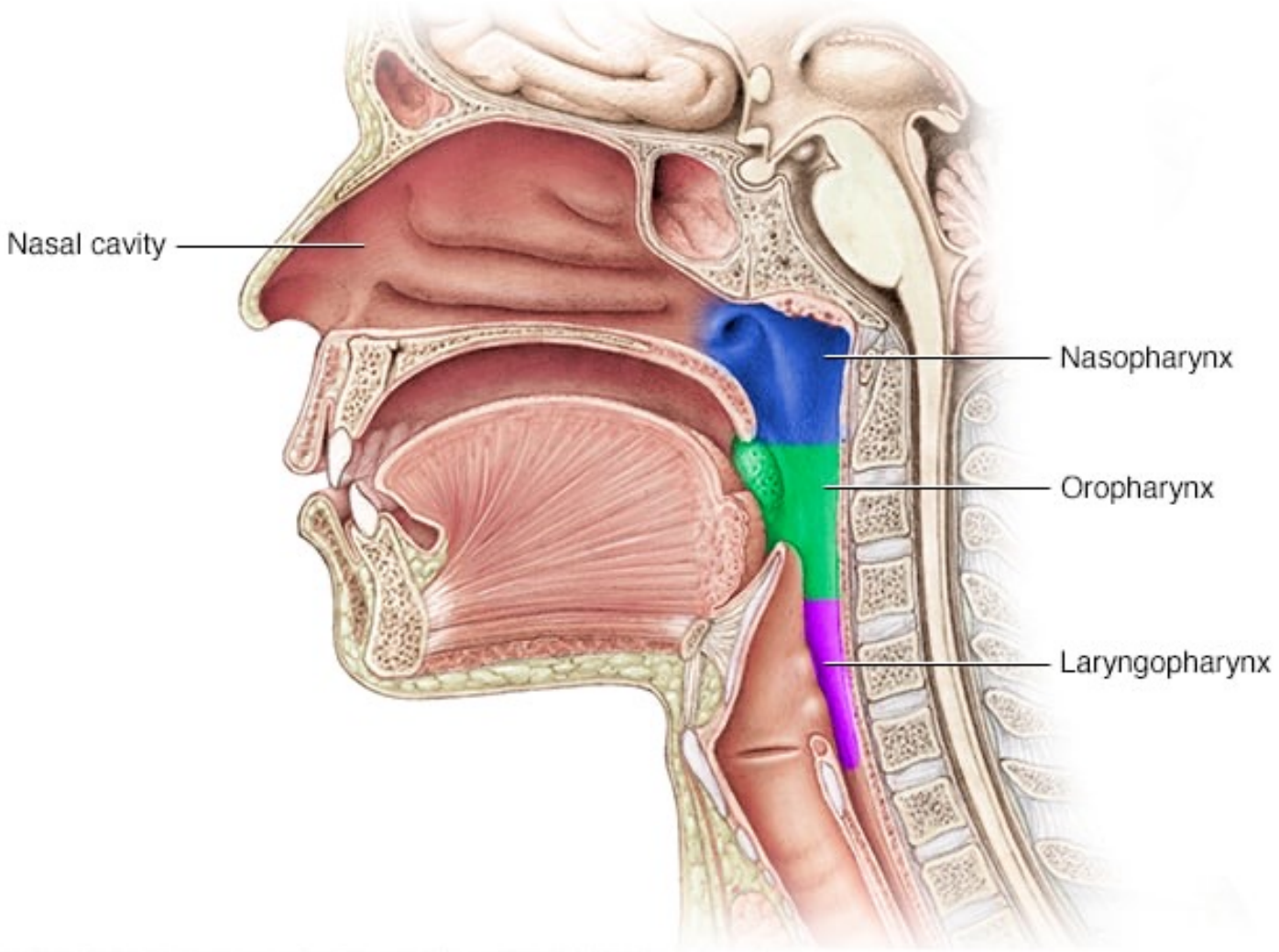


The Pharynx:

Pharynx – nasal cavity opens into a passage through the glottis

Pharynx divided into 3 regions:

- **Nasopharynx** – serves as airway
  - Auditory (eustachian) tubes connecting middle ear cavity open into nasopharynx
- **Oropharynx** – passageway for air and food
  - changes to squamous epithelium
  - also contain set of tonsils
- **Laryngopharynx** – continues route for ingested food and air until divergence of digestive and respiratory systems
  - Anteriorly laryngopharynx opens into larynx and posteriorly enters oesophagus

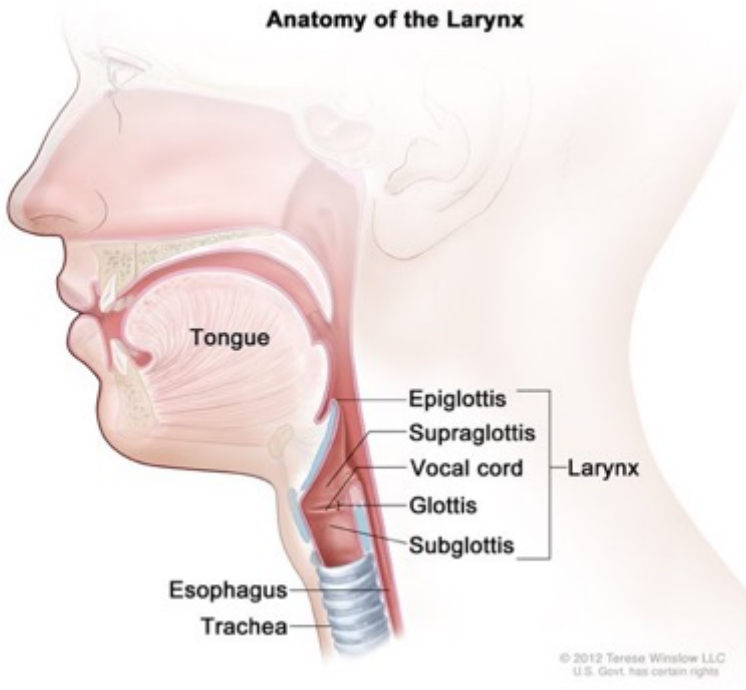


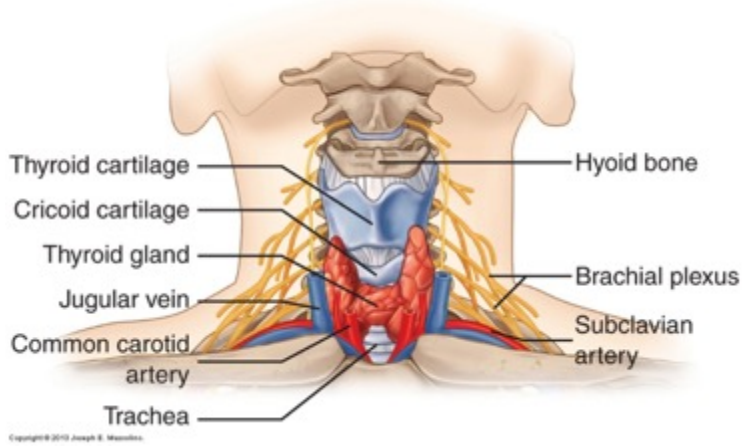
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The Larynx:

Larynx – cartilaginous structure which helps regulate volume of air entering and leaving lungs

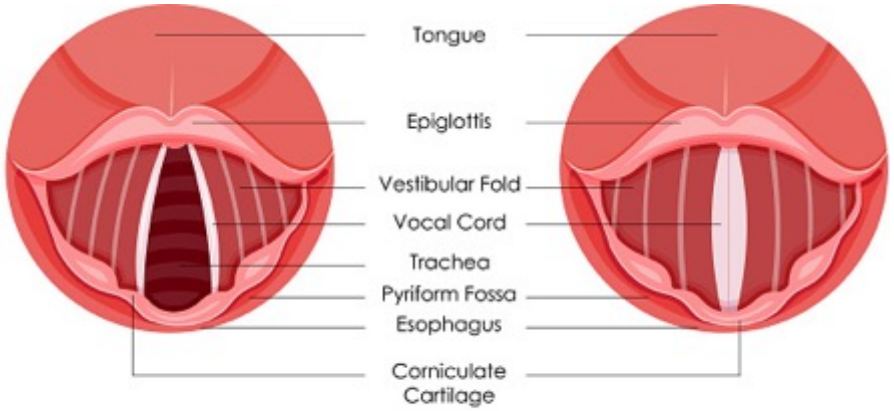
- Continuous with trachea inferiorly and pharynx superiorly
- 3 large cartilage pieces (thyroid, epiglottis, cricoid)
- ‘Adams apple’ - part of thyroid cartilage
- Epiglottis – flexible piece of elastic cartilage covers opening of trachea, to prevent aspiration



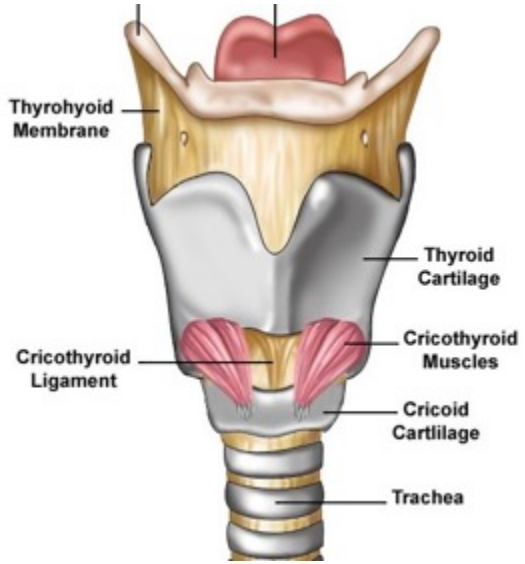


The Larynx

- Cartilaginous skeleton with ligaments and muscles that move and stabilise the surrounding mucus membrane.
- Formed of 9 cartilages:
  - Thyroid : protective shield. Largest cartilage and forms the adam's apple in men
  - Cricoid: encircles the trachea completely, made up of the arch (ant) and the lamina (post)
  - Epiglottis : leaf-shaped flap, attached to the internal surface of the thyroid cartilage
  - Arytenoid : pair of small, hard but flexible pyramid-shaped cartilages that sit over posterior cricoid.
  - Corniculate : pair of small elastic cone shaped cartilages that articulate with arytenoid
  - Cuneiform : two elongated fibrous pieces of yellow cartilage. Support the vocal folds and the lateral aspects of the epiglottis

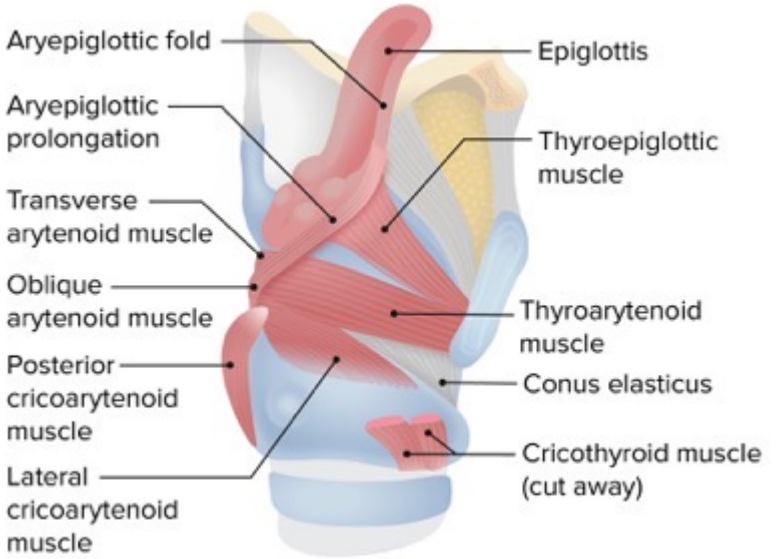


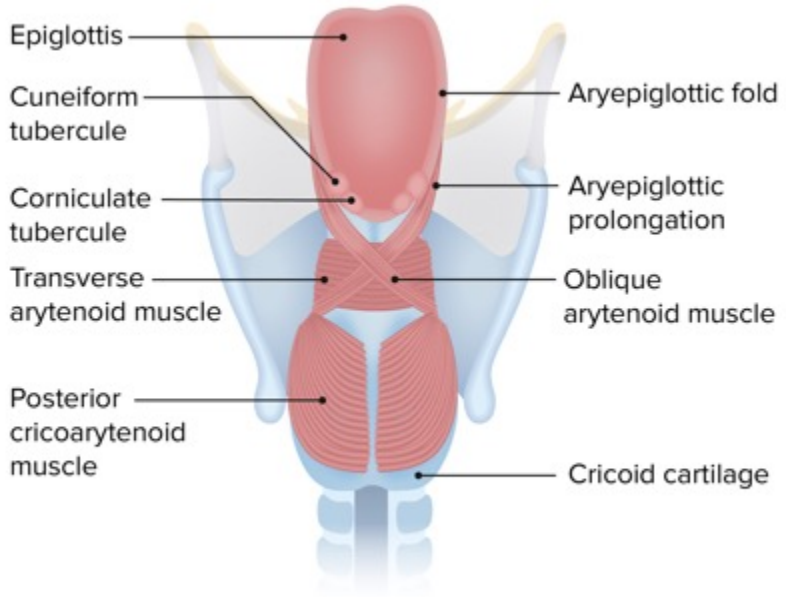
VOCAL CORD



Intrinsic muscles of the larynx

- Cricothyroid mm
- Posterior cricoarytenoid mm
- Lateral cricoarytenoid mm
- Thyroarytenoid mm
- Aryepiglottic mm
- Arytenoid mm

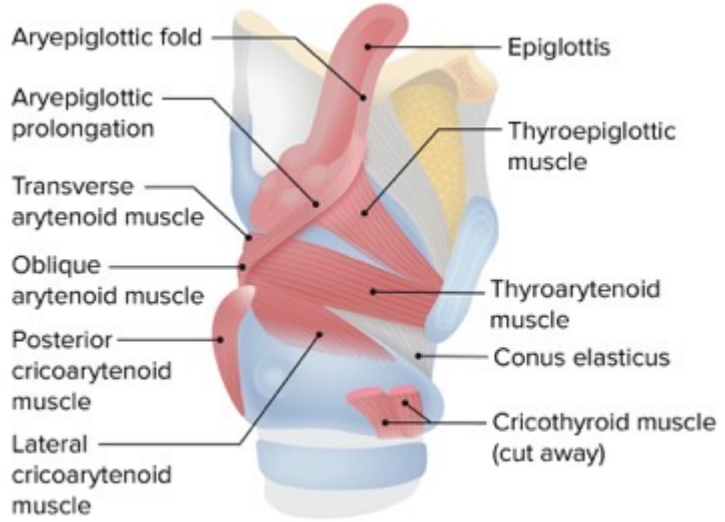


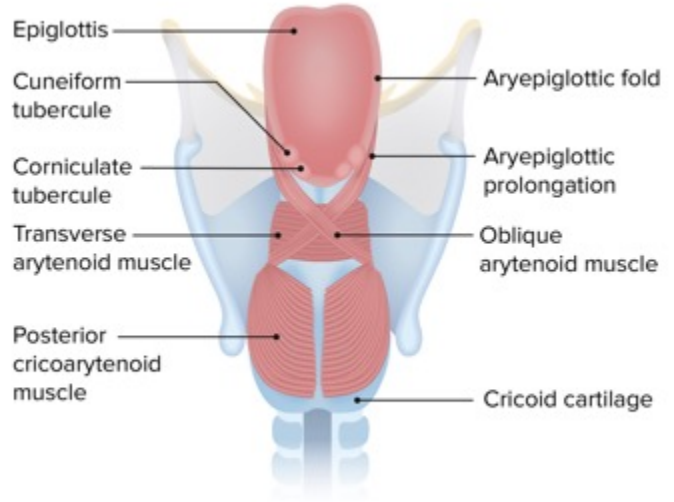


External muscles of the larynx

Paired and allow movement of the larynx

- Thyrohyoid mm
- Sternothyroid mm
- Inferior pharyngeal constrictor
- Stylopharyngeus mm
- Palatopharyngeus mm





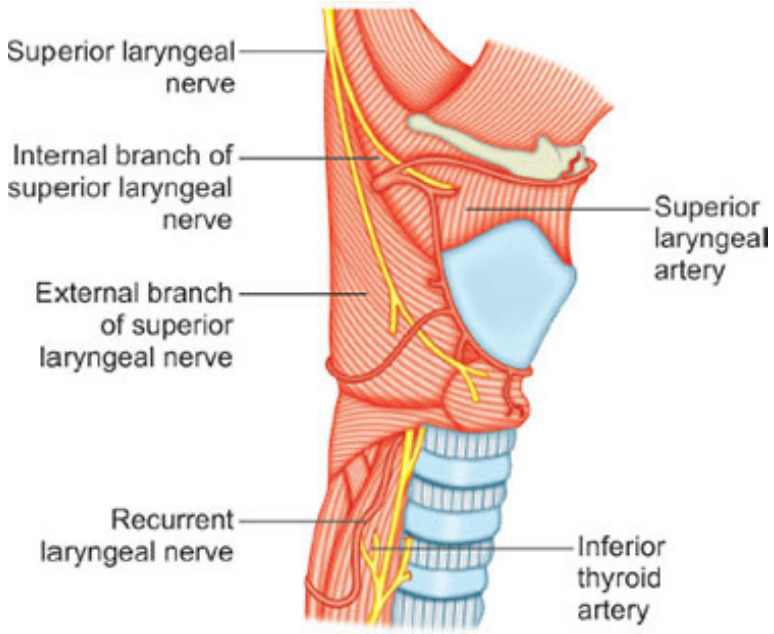
Innervation and blood supply

Innervation:

- Larynx receives **innervation** by the inferior laryngeal nerve, the recurrent laryngeal nerve and the superior laryngeal nerve
- Superior laryngeal nerve
  - A branch of the vagus nerve.
  - Divides into 2 braches at hyoid bone : internal and external
- Inferior laryngeal nerve
  - Principal nerve responsible for the innervation of all intrinsic mm bar the cricothyroid mm.
- Left recurrent nerve
  - Arises from vagus nerve, loops around aortic arch and up to the larynx
- Right recurrent nerve
  - Branch of vagus nerve at base of the neck , loops around the subclavian artery

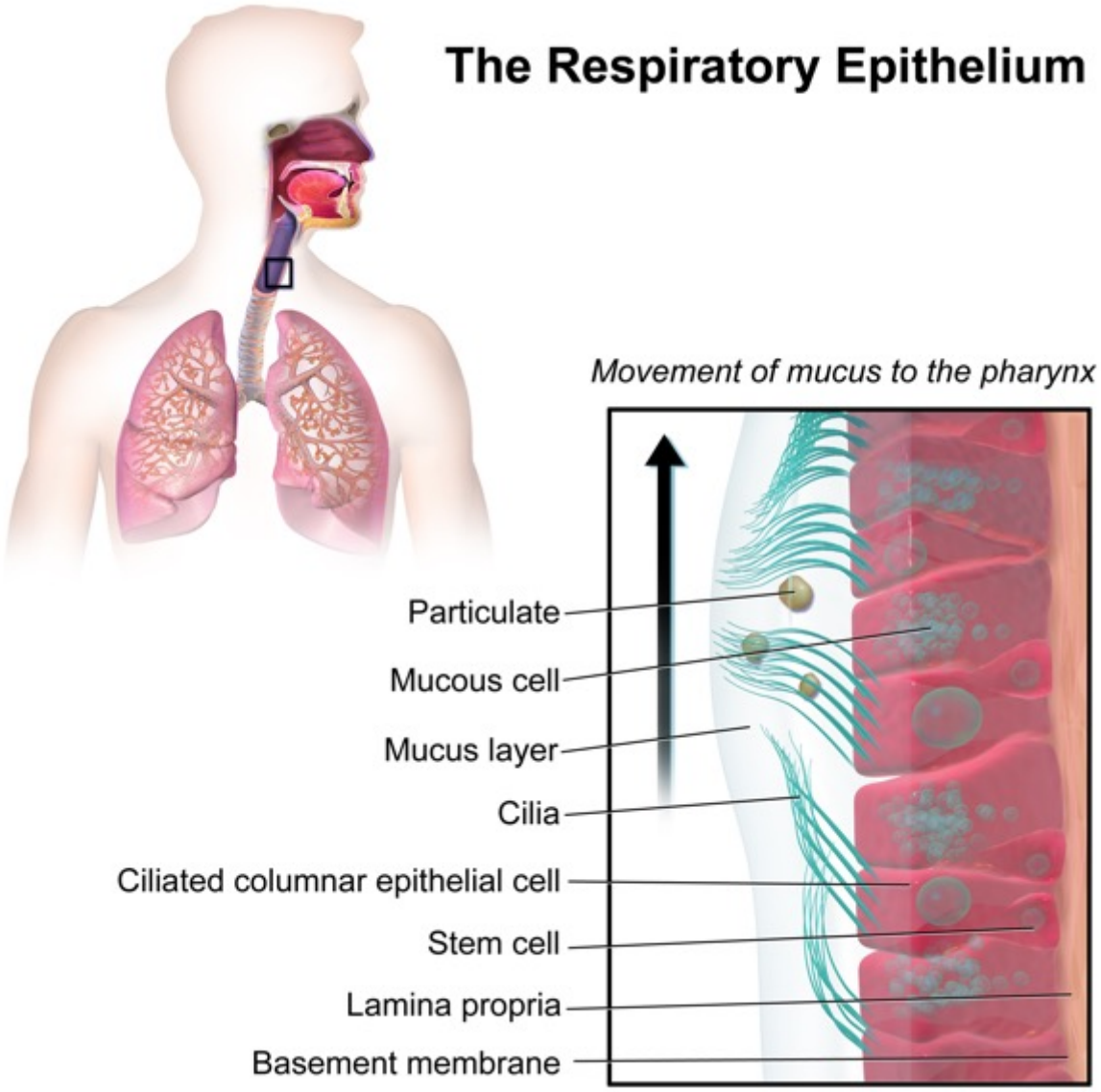
Blood supply

- Superior and inferior laryngeal arteries
- Superior :
  - Epiglottis
  - Supraglottic region
  - Superior vocal cords
- Inferior :
  - Subglottic region
  - Inferior vocal cords



Mucociliary transport system

- Primary defence mechanism of the lungs
- Three functional components:
  - Cilia
  - Mucus
  - Aqueous layer
  - Think of a toothbrush!



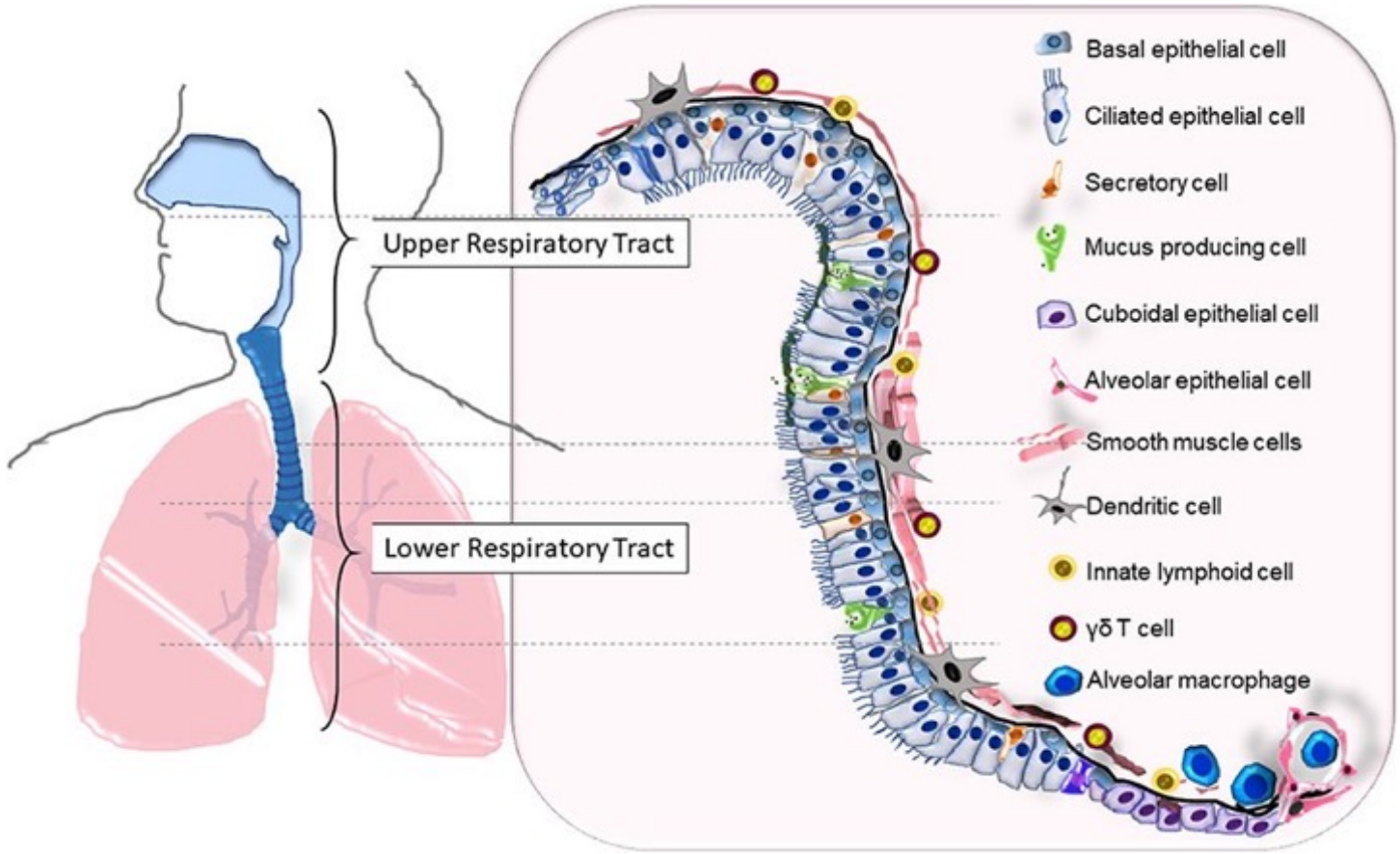


Immune cells of the airways

- Cellular components :
  - Phagocytes : neutrophils or macrophages
  - Natural killer cells
  - Basophils
  - Mast cells
  - Eoinophils
  - Epithelial cells of the airway
    - The epithelial cells not only provide a barrier but also contribute to the immune function

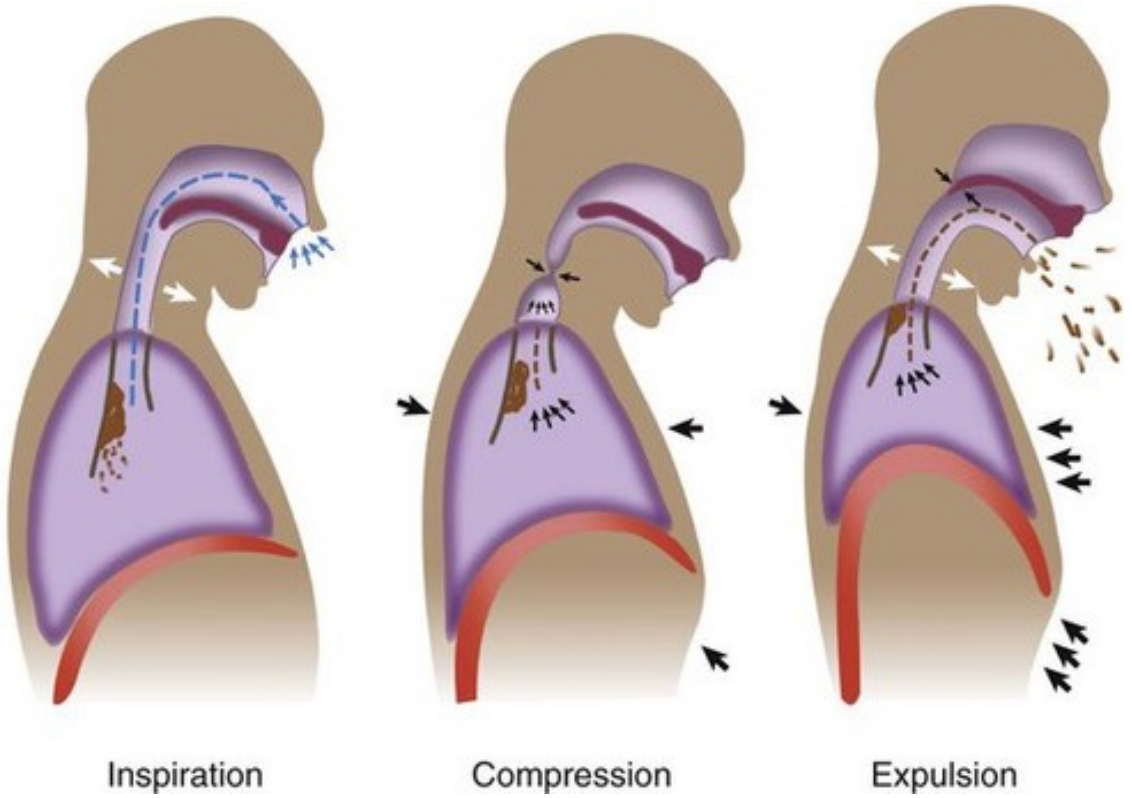
Immune response in the airways

- The airway epithelium senses bacterial exposure and responds
- Phagocytes, dendritic cells and epithelial cells bind too conserved molecules
- This starts an inflammatory / chemotactic or antimicrobial cascade
- Any changes to this system create dysfunction and thus illness e.g. Asthma, pneumonia, COPD, CF etc



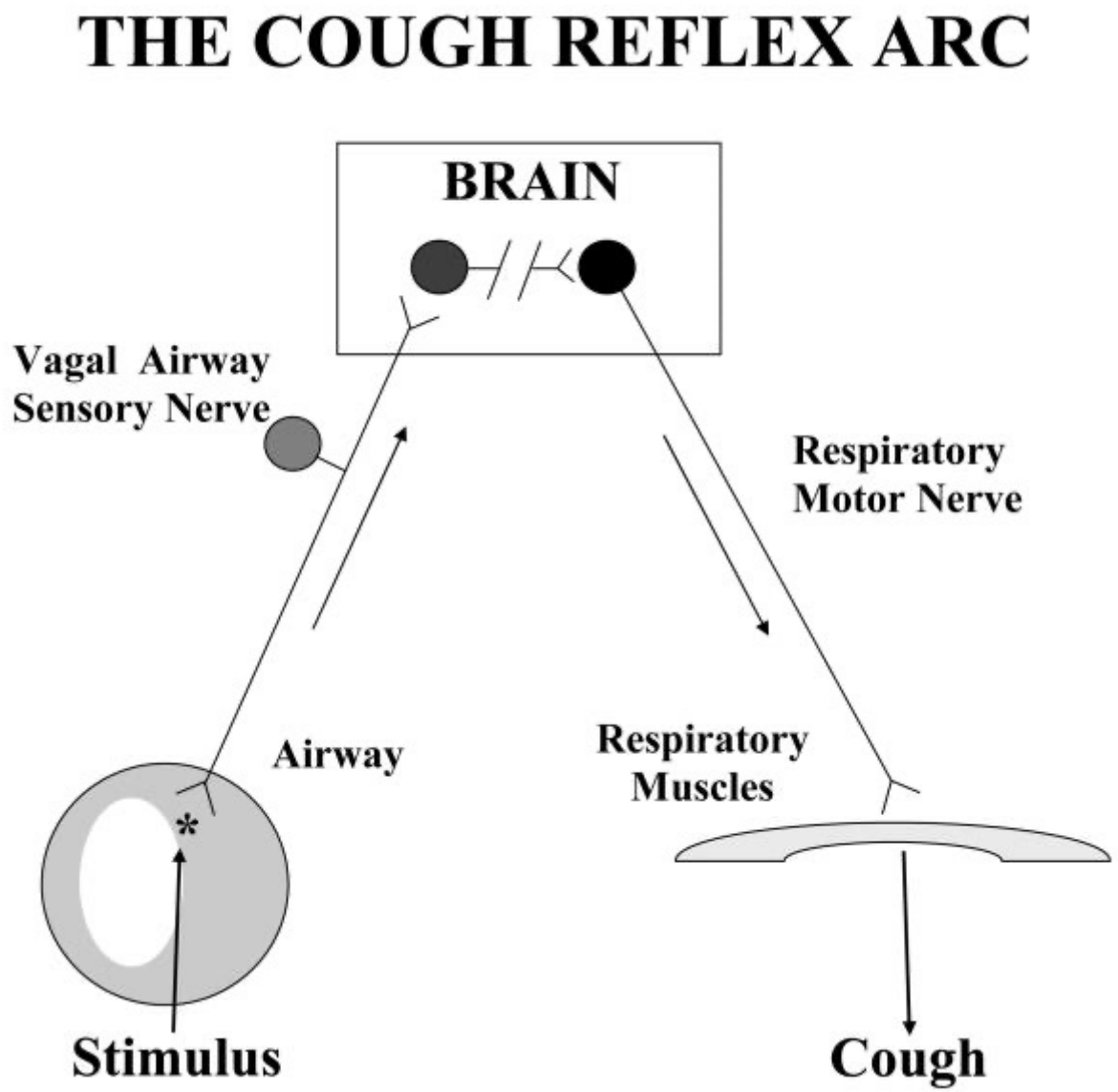
Coughing

- Important component of lung health
- Dependent on airway characteristics and integrity of neurophysiological pathway
- 3 phases :
  - **Inspiration** : Generates the volume necessary for an effective cough
  - **Compressive** : closure of the larynx combined with contraction of muscles of chest wall, diaphragm and abdominal wall. Rapid rise in intrathoracic pressure
  - **Expiratory :** The glottis opens, resulting in high expiratory airflow. Airway compression occurs. The high flow dislodges mucus from the airways
- With rapid coughing intrathoracic pressure could rise as high as 300mmHg and expulsion could be as fast as 800km/hr



Coughing

- Cough occurs through the stimulation of a reflex arc:
  - **Afferent pathway** : nerve fibres located in the epithelium of the upper airways travel via vagus nerve to the medulla
  - **Central pathway (cough centre)** : coordinating region for coughing is located in the upper brain stem and pons
  - **Efferent pathway** : impulses travel from the cough centre via the vagus, phrenic and spinal motor nerves to the diaphragm, abdominal wall, mm and the larynx.



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